



12.0.1 NIELIT 2016 MAR Scientist C - Section B: 17



A ladder 13 feet long rests against the side of a house. The bottom of the ladder slides away from the house at a rate of 0.5 ft/s. How fast is the top of the ladder sliding down the wall when the bottom of the ladder is 5 feet from the house?

A.

$$\frac{5}{24} \text{ ft/s}$$

B.

$$\frac{5}{12} \text{ ft/s}$$

C.

$$-\frac{5}{24} \text{ ft/s}$$

$$\text{D. } -\frac{5}{12} \text{ ft/s}$$

nielit2016mar-scientistc engineering-mathematics calculus

Answer key

12.0.2 NIELIT 2016 MAR Scientist C - Section B: 13



$$\lim_{x \rightarrow 0} \frac{1}{x^6} \int_0^{x^2} \frac{t^2 dt}{t^6 + 1} = ?$$

$$\text{A. } 1/4$$

$$\text{B. } 1/3$$

$$\text{C. } 1/2$$

$$\text{D. } 1$$

nielit2016mar-scientistc calculus

Answer key

12.0.3 NIELIT 2016 MAR Scientist C - Section B: 12



$$\lim_{x \rightarrow a} \frac{1}{x^2 - a^2} \int_a^x \sin(t^2) dt = ?$$

$$\text{A. } 2a \sin(a^2)$$

$$\text{B. } 2a$$

$$\text{C. } \sin(a^2)$$

$$\text{D. None of the above}$$

nielit2016mar-scientistc engineering-mathematics calculus

Answer key

12.0.4 NIELIT 2023 Feb Scientist D | Section D | Question: 98



Given $A = \sin^2 \theta + \cos^4 \theta$ then for all real values of θ :

$$\text{A. } 1 \leq A \leq 2$$

$$\text{B. } \frac{3}{4} \leq A \leq 1$$

$$\text{C. } \frac{13}{16} \leq A \leq 1$$

$$\text{D. } \frac{3}{4} \leq A \leq \frac{13}{16}$$

nielit2023feb-scientistd calculus

12.1.1 Continuity: NIELIT 2017 DEC Scientific Assistant A - Section B: 10



The function $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ is :

- A. Continuous and differentiable
- B. Continuous but not differentiable
- C. Differentiable but not continuous
- D. Neither continuous nor differentiable

nielit2017dec-assistanta engineering-mathematics calculus continuity

Answer key

12.1.2 Continuity: NIELIT2017 STA-set-c-119



The function $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ is:

- A. Continuous and Differentiable
- B. Continuous but not Differentiable
- C. Differentiable but not Continuous
- D. Neither Continuous nor Differentiable

nielit-july-2017 engineering-mathematics calculus continuity

Answer key

12.2 Coordinate Transformation (2)

12.2.1 Coordinate Transformation: NIELIT 2021 Dec Scientist B - Section B: 102

On changing to spherical co-ordinates, the integral $\iiint_V dy dx dz$, where V is the volume of the hemisphere $x^2 + y^2 + z^2 = a^2$, is equivalent to the integral _____.



- A. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin \theta dr d\theta d\phi$
- B. $\int_0^{2\pi} \int_0^{\pi} \int_0^a r^2 \sin \theta dr d\theta d\phi$
- C. $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin \theta dr d\theta d\phi$
- D. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \cos \theta dr d\theta d\phi$

nielit2021dec-scientistb calculus coordinate-transformation integrals

12.2.2 Coordinate Transformation: NIELIT 2021 Dec Scientist B - Section B: 35

On changing to spherical co-ordinates, the integral $\iiint_V dy dx dz$, where V is the volume of the hemisphere $x^2 + y^2 + z^2 = a^2$, is equivalent to the integral _____.



- A. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin \theta \, dr \, d\theta \, d\phi$
- B. $\int_0^{2\pi} \int_0^{\pi} \int_0^a r^2 \sin \theta \, dr \, d\theta \, d\phi$
- C. $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin \theta \, dr \, d\theta \, d\phi$
- D. $\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^a r^2 \cos \theta \, dr \, d\theta \, d\phi$

nielit-2021-it-dec-scientistb calculus integral coordinate-transformation

12.3

Definite Integral (4)

12.3.1 Definite Integral: NIELIT 2016 MAR Scientist B - Section B: 11



What is the derivative w.r.t x of the function given by

$$\Phi(x) = \int_0^{x^2} \sqrt{t} \, dt,$$

- A. $2x^2$ B. $\sqrt{x}$ C. 0 D. 1

nielit2016mar-scientistb engineering-mathematics calculus integration definite-integral

Answer key

12.3.2 Definite Integral: NIELIT 2016 MAR Scientist B - Section B: 9



The value of improper integral

$$\int_0^1 x \ln x \, dx = ?$$

- A. $1/4$ B. 0 C. $-1/4$ D. 1

nielit2016mar-scientistb engineering-mathematics calculus integration definite-integral

Answer key

12.3.3 Definite Integral: NIELIT 2016 MAR Scientist C - Section B: 11



$$\int_0^{\frac{\pi}{2}} \sin^7 \theta \cos^4 \theta \, d\theta = ?$$

- A. $16/1155$ B. $16/385$ C. $16\pi/385$ D. $8\pi/385$

nielit2016mar-scientistc engineering-mathematics calculus definite-integral

Answer key 

12.3.4 Definite Integral: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 19

The value of the Integral $I = \int_0^{\pi/2} x^2 \sin x dx$ is



- A. $(x+2)/2$ B. $2/(\pi-2)$ C. $\pi-2$ D. $\pi+2$

nielit2017oct-assistanta-cs engineering-mathematics calculus definite-integral

Answer key 

12.4 Differential Equation (1)

12.4.1 Differential Equation: NIELIT 2016 MAR Scientist B - Section B: 15



Differential equation, $\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$ will have a solution of the form

- A. $(C_1 + C_2t)e^{-5t}$ B. C_1e^{-2t}
C. $C_1e^{-5t} + C_2e^{5t}$ D. $C_1e^{-5t} + C_2e^{2t}$
where C_1 and C_2 are constants.

nielit2016mar-scientistb non-gatecse differential-equation

12.5 Functions (2)

12.5.1 Functions: NIELIT 2022 April Scientist B | Section B | Question: 62



The function $f(x) = x(x+3)e^{-\frac{x}{2}}$ satisfies all the conditions of Rolle's theorem in $[-3, 0]$. The value of c is:

- A. -3 B. -2 C. 3 D. 0

nielit2022apr-scientistb calculus functions

Answer key 

12.5.2 Functions: NIELIT 2023 Feb Scientist C | Section D | Question: 105



The value of $f(x, y) = \left(\frac{\sqrt[4]{x^3y} - \sqrt[4]{xy^3}}{\sqrt{y} - \sqrt{x}} + \frac{1 + \sqrt[2]{xy}}{\sqrt[4]{xy}} \right)^{-2} \cdot \left(1 + 2\sqrt{\frac{y}{x}} + \frac{y}{x} \right)^{\frac{1}{2}}$ when $x = 9, y = 0.04$:

- A. 0.64 B. 0.46 C. 0.32 D. 0.11

nielit2023feb-scientistc functions calculus

12.6 Integral (1)

12.6.1 Integral: NIELIT 2021 Dec Scientist A - Section B: 116



$\iint \frac{xy}{\sqrt{1-y^2}} dx dy$. Over the positive quadrant of the circle $x^2 + y^2 = 1$ is

- A. $\frac{1}{6}$ B. $\frac{2}{3}$ C. $\frac{5}{6}$ D. $\frac{5}{3}$

nielit2021dec-scientista calculus integral geometry

12.7 Limits (2)

12.7.1 Limits: NIELIT 2016 MAR Scientist B - Section B: 13



$\lim_{x \rightarrow 0} \frac{(1 - \cos x)}{2}$ is equal to

- A. 0 B. 1 C. $1/3$ D. $1/2$

nielit2016mar-scientistb engineering-mathematics calculus limits

Answer key

12.7.2 Limits: NIELIT 2016 MAR Scientist C - Section B: 10



$\lim_{x \rightarrow 0} \frac{x^3 + x^2 - 5x - 2}{2x^3 - 7x^2 + 4x + 4} = ?$

- A. -0.5 B. (0.5) C. ∞ D. None of the above

nielit2016mar-scientistc engineering-mathematics calculus limits

Answer key

12.8 Maxima Minima (8)

12.8.1 Maxima Minima: NIELIT 2016 DEC Scientist B (CS) - Section B: 26



Consider the function $f(x) = \sin(x)$ in the interval $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$. The number and location(s) of the minima of this function are:

- A. One, at $\frac{\pi}{2}$ B. One, at $\frac{3\pi}{2}$
C. Two, at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$ D. Two, at $\frac{\pi}{4}$ and $\frac{3\pi}{2}$

nielit2016dec-scientistb-cs engineering-mathematics calculus maxima-minima

Answer key

12.8.2 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 10



Maxima and minimum of the function

$f(x) = 2x^3 - 15x^2 + 36x + 10$ occur; respectively at

- A. $x = 3$ and $x = 2$ B. $x = 1$ and $x = 3$ C. $x = 2$ and $x = 3$ D. $x = 3$ and $x = 4$

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key

12.8.3 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 14



The minimum value of $|x^2 - 5x + 2|$ is

- A. -5 B. 0 C. -1 D. -2

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key

12.8.4 Maxima Minima: NIELIT 2016 MAR Scientist B - Section B: 5



The greatest and the least value of

$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 1$ in $[0, 2]$ are

- A. $0, 8$ B. $0, -8$ C. $1, 8$ D. $1, -8$

nielit2016mar-scientistb engineering-mathematics calculus maxima-minima

Answer key

12.8.5 Maxima Minima: NIELIT 2016 MAR Scientist C - Section B: 2



The function $f(x) = x^5 - 5x^4 + 5x^3 - 1$ has

- A. one minima and two maxima B. two minima and one maxima
C. two minima and two maxima D. one minima and one maxima

nielit2016mar-scientistc calculus maxima-minima

Answer key

12.8.6 Maxima Minima: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 18



What is the maximum value of the function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$?

- A. 6 B. 10 C. 12 D. $5, 5$

nielit2017oct-assistanta-cs engineering-mathematics calculus maxima-minima

Answer key

12.8.7 Maxima Minima: NIELIT 2021 Dec Scientist A - Section B: 70



With usual notations, the properties of maxima and minima under various conditions are _____.

I

(P) Maxima

(Q) Minima

(R) Saddle Point

(S) Case of failure

II

(i) $rt - s^2 = 0$

(ii) $rt - s^2 < 0$

(iii) $rt - s^2 > 0, r > 0$

(iv) $rt - s^2 > 0, r < 0$

- A. P – i, Q – iii, R – iv, S – ii
- B. P – ii, Q – i, R – iii, S – iv
- C. P – iii, Q – iv, R – ii, S – i
- D. P – iv, Q – iii, R – ii, S – i

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12.8.8 Maxima Minima: NIELIT 2022 April Scientist B | Section B | Question: 112



The function $f(x, y) = x^2 + y^2 + 6x + 12$ has :

- A. minimum value, -3 and maximum value, 0 .
- B. only minimum value, 3 .
- C. only maximum value, 12 .
- D. neither maxima nor minima.

nielit2022apr-scientistb calculus maxima-minima functions

12.9

Numerical Methods (1)

12.9.1 Numerical Methods: NIELIT 2022 April Scientist B | Section B | Question: 88



The secant method formula for finding the square root of a real number R from the equation $x^2 - R = 0$ is:

A. $x_{i+1} = \frac{x_i x_{i-1}}{x_i + x_{i-1}}$

C. $x_{i+1} = \frac{1}{2} \left(x_i + \frac{R}{x_i} \right)$

B. $x_{i+1} = \frac{x_i x_{i-1} + R}{x_i + x_{i-1}}$

D. $x_{i+1} = \frac{2x_i^2 + x_i x_{i-1} - R}{x_i + x_{i-1}}$

nielit2022apr-scientistb calculus numerical-methods

Answer key

12.10

Partial Order (2)

12.10.1 Partial Order: NIELIT 2021 Dec Scientist B - Section B: 41



If $z = \cos\left(\frac{x}{y}\right) + \sin\left(\frac{x}{y}\right)$, then $xz_x + yz_y$ is equal to _____.

A. z B. $2z$ C. $4z$

D. 0

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12.10.2 Partial Order: NIELIT 2021 Dec Scientist B - Section B: 95

If $z = \cos\left(\frac{x}{y}\right) + \sin\left(\frac{x}{y}\right)$, then $xz_x + yz_y$ is equal to _____.

A. z B. $2z$ C. $4z$

D. 0

nielit2021dec-scientistb calculus partial-order functions

12.11**Quadratic Equations (1)****12.11.1 Quadratic Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 95**

Find the roots of equation $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x} : a + b \neq 0$.

A. a, b B. $-a, -b$ C. $2a, 3b$ D. $3a, 2b$

nielit2023feb-scientistc quadratic-equations algebra

Answer Keys

12.0.1	C
12.1.2	D
12.3.3	A
12.6.1	A
12.8.3	B
12.8.8	B

12.0.2	B
12.2.1	A
12.3.4	C
12.7.1	A
12.8.4	D
12.9.1	C

12.0.3	D
12.2.2	A
12.4.1	A
12.7.2	A
12.8.5	A
12.10.1	D

12.0.4	B
12.3.1	A
12.5.1	B
12.8.1	D
12.8.6	B
12.10.2	D

12.1.1	D
12.3.2	C
12.5.2	A
12.8.2	C
12.8.7	D
12.11.1	B